

Bridging Classical and Data-Driven Transport Modelling

New Approaches to Traffic Prediction on the UK Road Network

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Thesis structure

1	Introduction	Motivation, research problem, research questions and contributions	
2	Background and literature review	Four-step travel demand model, data-driven prediction and estimation, machine learning within the four-step model, integrated demand-to-flow modelling	
3	Methodology and data	Shared prediction task, road-network, socioeconomic and traffic data, evaluation design	
4	Mukara	CNN and graph attention network on 498 directed trunk-road segments	RQ1 · RQ3
5	DeepDemand	Theory-informed OD representation on 5,088 monitored edges of the national driving network	RQ2 · RQ3
6	Conclusions	Answers to the research questions, contributions, scope and further work	

Both models share one inference constraint: observed traffic counts supervise fitting and evaluation but are never prediction inputs.

CHAPTER 1

Introduction

Motivation · research problem · research questions and contributions

Background and motivation

- Link-level traffic volumes inform infrastructure investment, congestion management, emissions regulation and public-health assessment.
- Road transport dominates travel. Cars, vans and taxis carried **86%** of passenger-kilometres in Great Britain in 2022.
- Much recent deep learning forecasts traffic minutes to hours ahead from **recent sensor observations**.
- Planning faces a different information constraint. Many roads have no measurement history, and traffic must be related to population, employment, land use and network structure.
- Empirical setting: interurban highway traffic on England's Strategic Road Network (SRN), with socioeconomic inputs for England and Wales.

Three requirements used to assess the models

Accuracy

Agreement with held-out traffic counts

Spatial generalisation

Prediction in regions excluded from fitting

Interpretability

Examination of learned demand-related relationships

Related but distinct. Random hold-out accuracy does not establish geographical transfer, and accuracy alone does not establish what learned quantities mean (Figure 1.1).

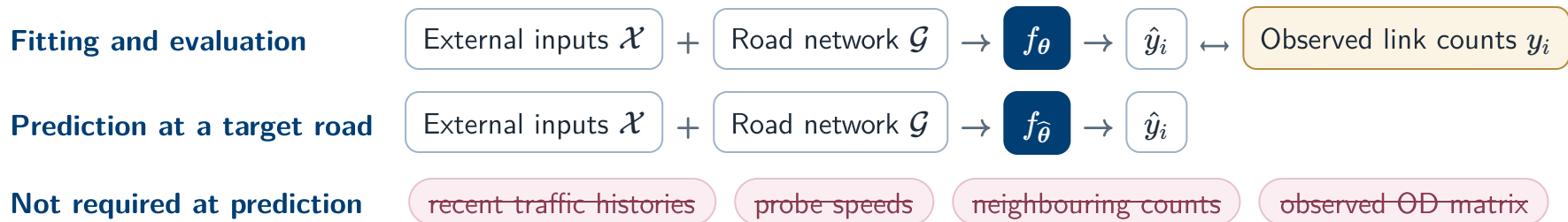
Two information structures

	Four-step travel demand model (TDM)	Sensor-history deep learning
Target	Trip totals, OD movements, modal shares and assigned link volumes	Future traffic states, minutes to hours ahead
Inputs	Socioeconomic and land-use characteristics, travel services, costs and network	Recent traffic observations at the target and neighbouring sensors
Representation	Explicit generation, distribution, mode-choice and assignment relationships	Learned spatial and temporal dependencies in observed traffic
Evaluation	Calibration and validation at several intermediate and network levels	Held-out temporal predictions against measured traffic states
Main constraint	Data, assumptions and local calibration across several components	Traffic-state observations are required when predictions are made

After Table 1.1

Other approaches sit between the two: external-attribute estimators, probe-based reconstruction, machine learning inside single TDM stages, joint estimation and simulator surrogates. Each addresses part of the demand-to-flow chain (Chapter 2).

Research problem: supervision versus inference



- Existing methods address portions of the demand-to-flow chain, or rely on traffic-state observations or intermediate demand quantities when predicting.
- Less developed setting: link-count supervision, with inference from external socioeconomic, land-use and network information only.
- How transport information is represented then matters as much as the choice of learning algorithm: flexible representation learning or explicit theory-informed structure.

Central hypothesis. External demand-related characteristics and network structure contain sufficient information to support useful link-level traffic estimation without traffic-state measurements at prediction time, and transport theory can guide the representation of these relationships in ways that support both prediction and interpretation.

Research questions and contributions

RQ1 · Mukara

Can highway traffic be estimated from external socioeconomic characteristics and road-network structure without using traffic observations as prediction inputs?

RQ2 · DeepDemand

Can a more explicit representation of OD relationships informed by travel-demand theory support prediction and interpretation at a finer road-segment resolution?

RQ3 · both models

How well do the models generalise geographically, and what can be learned from their fitted internal relationships?

Contributions

- 1. Observation-supervised external-input estimation.** Two models learn link volumes from counts, without traffic-state inputs or an observed OD matrix at inference.
- 2. Explicit test of geographical generalisation.** Region-held-out CV alongside random CV in both studies.
- 3. Theory-informed OD representation.** OD extraction, route screening, learned interaction and travel-time attenuation, implemented at national scale.

Shared task

Link traffic from socioeconomic context and network structure



Mukara

Learn spatial and network representations



finer network, more explicit demand structure

DeepDemand

Represent OD relationships explicitly

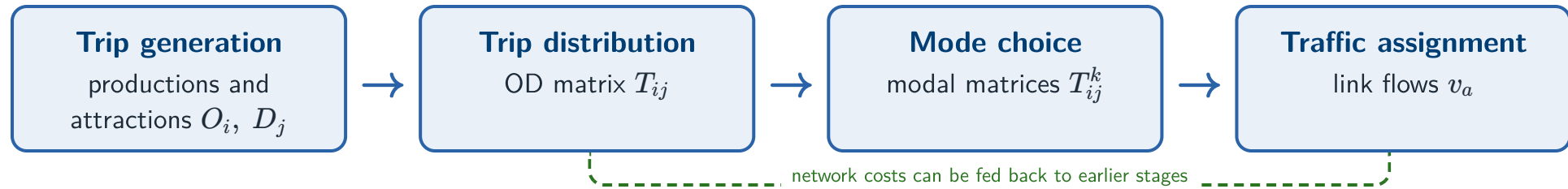
Methodological progression (Figure 1.2)

CHAPTER 2

Background and literature review

Four-step TDM · data-driven prediction and estimation · machine learning
within the TDM · integrated demand-to-flow modelling

The four-step travel demand model



Explicit, inspectable structure

- Gravity distribution $T_{ij} = A_i O_i B_j D_j f(c_{ij})$. Wilson's entropy derivation gives $f(c_{ij}) = \exp(-\beta c_{ij})$.
- Logit mode choice. User-equilibrium or stochastic user-equilibrium assignment with volume–delay functions.
- Each intermediate quantity can be compared with its own observations.

Calibration is not validation

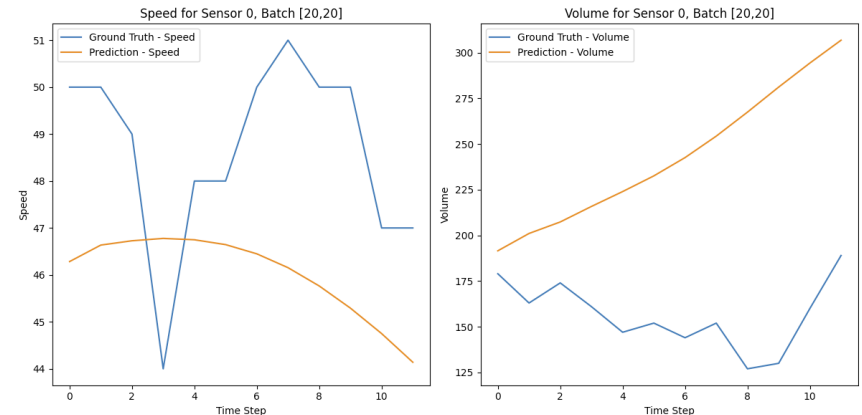
- OD matrices are adjusted until assigned flows reproduce counts. Agreement with the same counts measures in-sample calibration.
- Many OD and route-flow patterns reproduce the same link counts.
- Local fit is absorbed into a location-specific OD matrix, which cannot normally be transferred. More transferable parameters may be too coarse for link-level flows without local information.

Wherever a model is used beyond the network it was calibrated on, it needs validation in places whose counts did not contribute to fitting (Section 2.2).

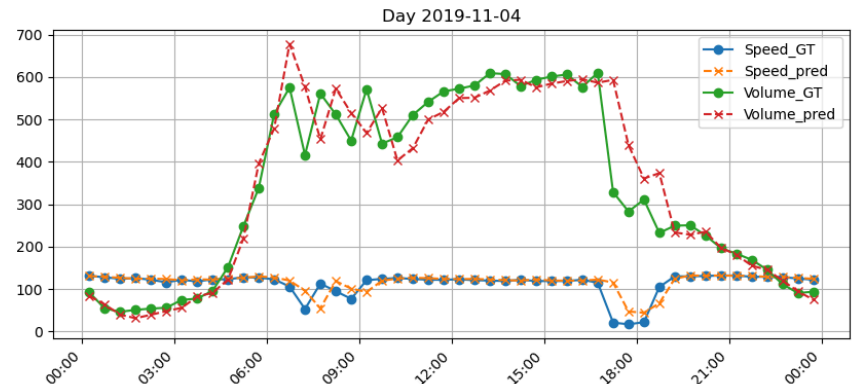
Case study: forecasting with ST-MetaNet

- ST-MetaNet on 498 monitored segments of England's trunk-road network. Speed and volume at 15-minute intervals, twelve steps in and twelve out, chronological 70/10/20 split.
- **Blind extrapolation.** Forecasts continue the recent trend smoothly and miss turning points.
- **Error accumulation.** In recursive decoding an early error is carried forward and grows over the horizon.
- **Lag.** A one-step variant (30-minute intervals, six hours of history) follows the daily profile but reproduces abrupt changes only after they enter the input history.

The forecast is conditional on an observed traffic state. Estimating typical volumes where no traffic history exists is a different task.



Twelve-step forecast: predicted volume rises while observed volume falls (Figure 2.5)



One-step forecast, 4 November 2019: the drop around 17:00 is reproduced one interval late (Figure 2.6)

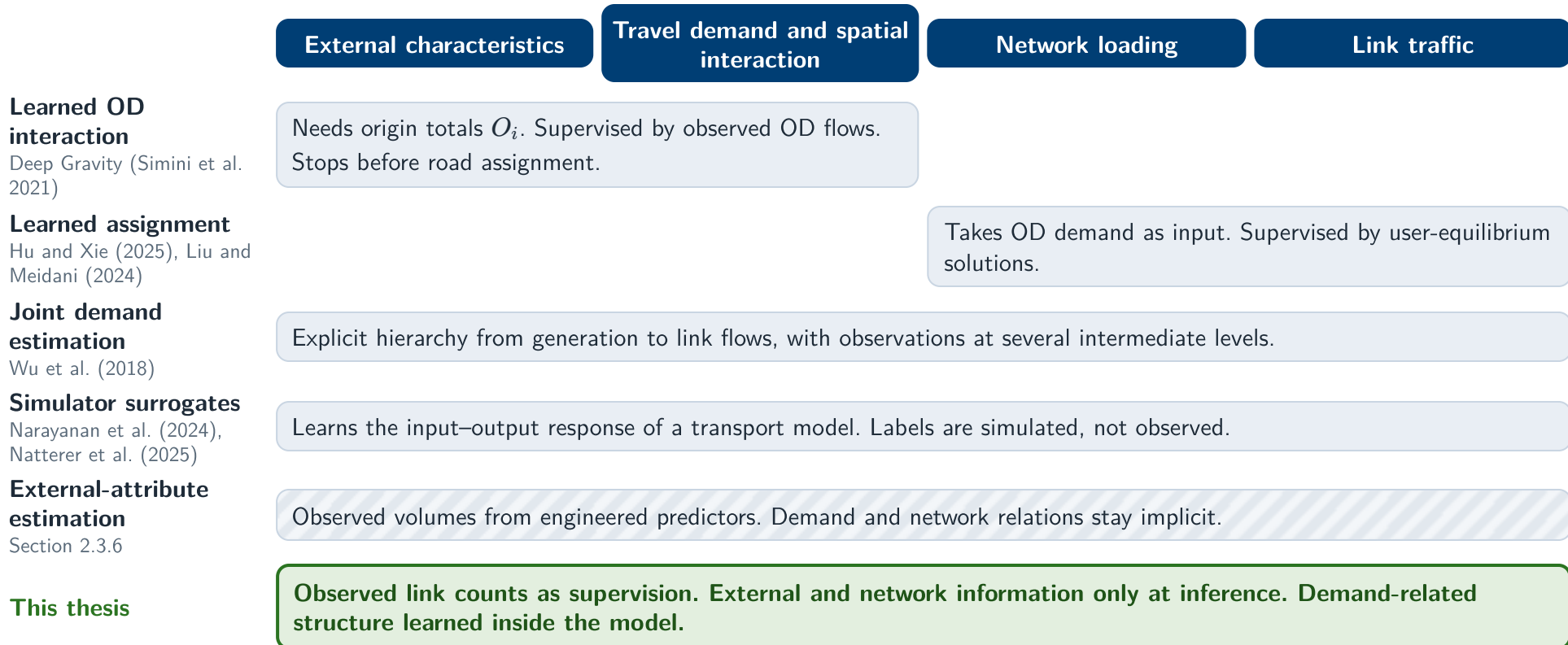
What information is available at prediction time?

Setting	Information required at inference	Representative studies
Short-term forecasting	Recent traffic history at the target and neighbouring sensors	DCRNN, STGCN, STGAT, ST-MetaNet
Reconstruction and kriging	Traffic observations at other locations	Zhang et al. (2020), IGNNK (Wu et al. 2021)
Probe-based estimation	Probe speeds, with no volume counter at the target	Sekuła et al. (2018), Hein et al. (2025)
External-attribute estimation	Roadway, socioeconomic and land-use characteristics only	Sfyridis and Agnolucci (2020, 2023), Das and Tsapakis (2020), van Strien and Grêt-Regamey (2024)

Rows run from the weakest to the strongest information constraint (Sections 2.3.6 and 2.3.7)

- External-only estimation has been demonstrated, mostly with engineered predictors attached to count points or network service areas.
- Network position carries information about volume (routing importance, Morley and Gulliver 2016), but that model also uses the measured AADT of the nearest major road.
- Evaluation ranges from model fit to geographical hold-out. Van Strien and Grêt-Regamey test transfer to countries excluded from calibration.
- Open: combining external characteristics with learned road-network relationships for link-level prediction, and testing transfer beyond the fitted locations.

Research gap: coverage of the demand-to-flow chain



Question. Can more of the demand-to-flow relationship be learned directly from observed link counts, when prediction uses only external characteristics and network structure rather than traffic measurements or a complete observed OD matrix?

After Table 2.3 and Figure 2.7

CHAPTER 3

Methodology and data

Shared prediction task · road-network, socioeconomic and traffic data ·
evaluation design

Shared prediction task

$$\hat{y}_i = f_{\theta}(\mathcal{X}, \mathcal{G}; i), \quad \hat{\theta} = \arg \min_{\theta} \frac{1}{|\mathcal{I}_{\text{train}}|} \sum_{i \in \mathcal{I}_{\text{train}}} \ell(y_i, f_{\theta}(\mathcal{X}, \mathcal{G}; i))$$

- $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is the directed road network. $\mathcal{X}$ collects the external socioeconomic, land-use and road-network inputs.
- Observed counts y_i enter only the training loss and the held-out evaluation.
- Mukara and DeepDemand differ in how they represent $\mathcal{X}$ and $\mathcal{G}$.

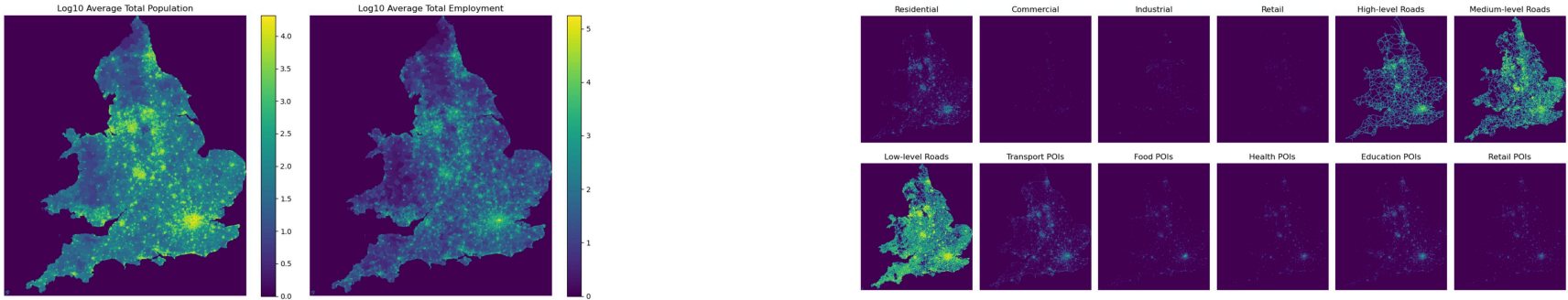
	Mukara	DeepDemand
Prediction unit	Directed trunk-road segment in a year, $i = (t, e)$	Directed road edge, $i = e$
Traffic period	2015–2022	2017–2024
Target	Mean weekday daily volume, per year	Median of annual AADT across valid years
Temporal exclusions	Weekends and bank holidays	Annual AADT from valid daily observations
Target edges	498	5,088
Traffic input at prediction	None	None

After Table 3.2. Both targets are in vehicles per day.

Data sources

Source	Coverage	Mukara	DeepDemand
National Highways SRN	England	Compact interurban trunk-road network	Monitored highway system holding the targets
OpenStreetMap	Great Britain	Land use, road context and POIs	National driving network, land use and POIs
ONS population estimates	England and Wales	Annual rasterised population	LSOA demographic features
Business Register and Employment Survey	England and Wales	Annual rasterised employment	LSOA employment features
Census 2021	England and Wales	–	Household composition, car or van availability
Deprivation data (2019)	England and Wales	–	Income and employment deprivation
Google Routes API	Mukara segments	Segment travel and geometric attributes	–
National Highways TRIS	England SRN	Targets on 498 directed segments	Targets on 5,088 directed edges

After Table 3.1. Small-area statistics use the Lower Layer Super Output Area (LSOA) geography.



Evaluation design and metrics

Random CV (five folds)

Held-out target edges spread across the network. Mukara keeps every year of an edge in the same fold. DeepDemand splits target edges 70/10/20 into training, validation and test sets.

Spatial CV (nine folds)

Each official region of England is held out in turn. The test region contributes no traffic counts to fitting. DeepDemand draws its validation set from the other eight regions.

- All models in a study share the same partitions. Results are means (standard deviations) across folds.
- Both schemes test spatial estimation. Neither tests future-year forecasting or the response to an intervention.

Metrics

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \quad (\text{vehicles per day})$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$\text{wMAPE} = 100 \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i y_i}$$

$$\text{GEH}_i = \sqrt{\frac{2(y_i - \hat{y}_i)^2}{y_i + \hat{y}_i}}, \quad \text{MGEH} = \frac{1}{n} \sum_i \text{GEH}_i$$

GEH depends on the flow period. The usual references of 5 and 10 apply to hourly flows (Backup).

CHAPTER 4

Mukara

RQ1 · Can highway traffic be estimated from external socioeconomic characteristics and road-network structure without using traffic observations as prediction inputs?

Mukara: objectives and data

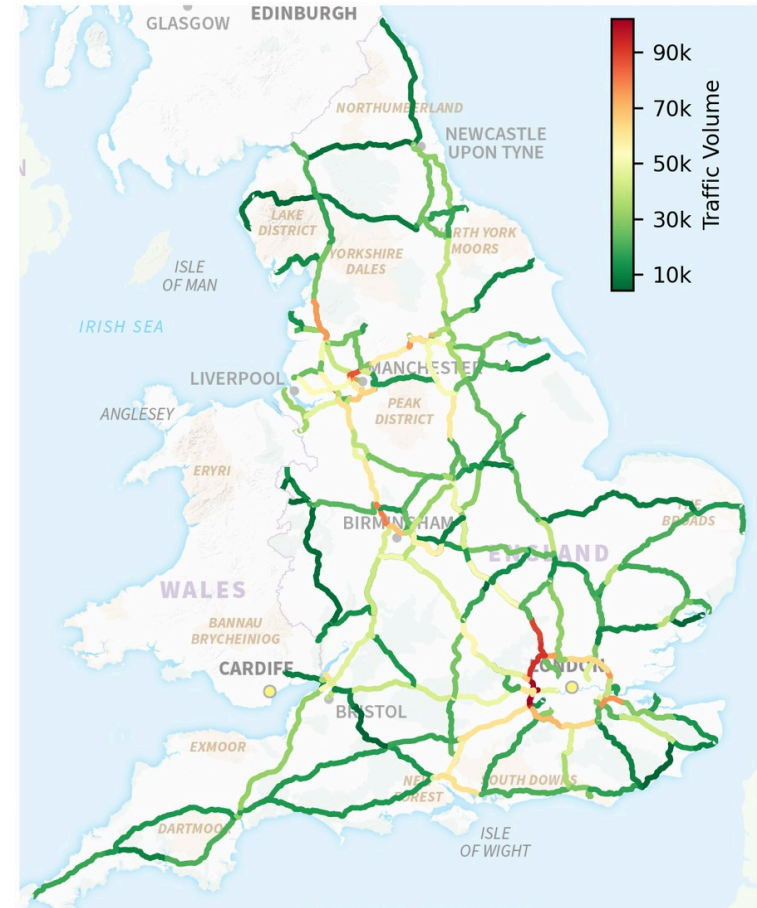
First test of the prediction setting, with deliberately limited theory structure: no OD pairs, OD matrix or route assignment. The name derives from the Japanese for “from nothing”, as no sensor history is an input.

Objectives

1. Estimate national highway traffic from socioeconomic characteristics and network structure.
2. Evaluate accuracy and spatial generalisation against baselines.
3. Examine the contributions of spatial context, network propagation and feature groups.

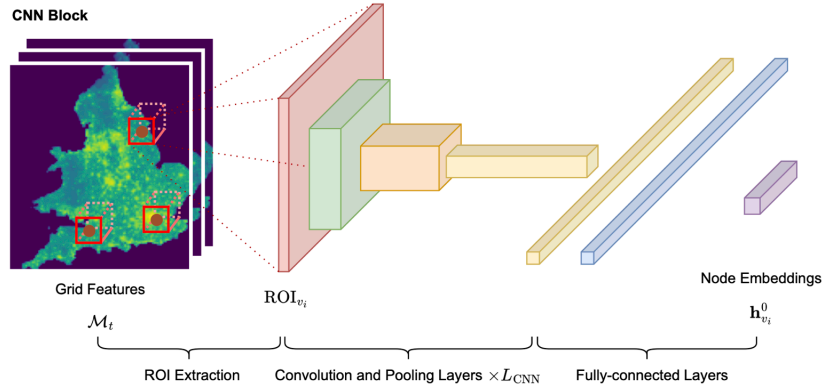
Data

- Trunk-road graph: 181 junction nodes, 498 directed edges (249 connections in both directions).
- One representative TRIS sensor per edge, chosen from 3,984 candidates by topology and data availability.
- 1 km grids: population and employment for 2015–2022, plus four land-use, three road-level and five POI channels.
- Five Google Routes API edge features: duration, distance, straight-line distance, speed, detour factor.
- Target: mean weekday daily traffic per edge and year (8×498 edge-year observations).

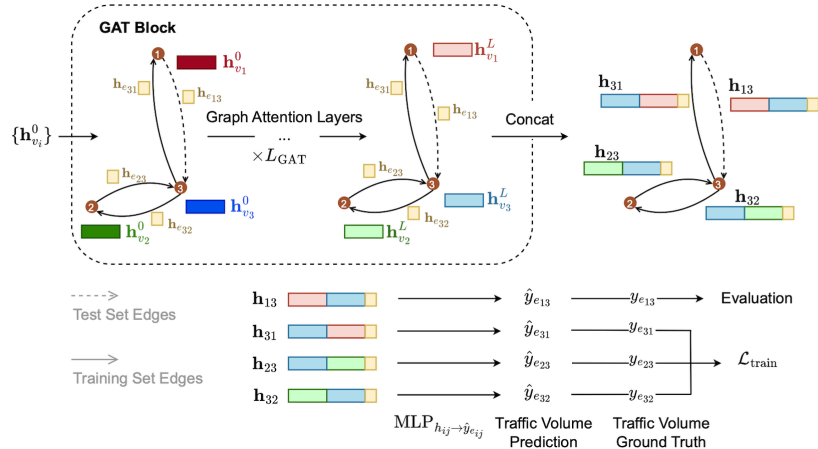


Mean weekday daily traffic, 2015–2022 (Figure 4.8)

Mukara: CNN encoder and edge-aware graph attention



CNN block: a region of interest around each junction becomes a node embedding (Figure 4.9)



GAT block: embeddings propagate over the trunk-road graph, then an MLP predicts each edge (Figure 4.10)

Node context from the grid

$$\mathbf{h}_{v_i}^{(0)} = \text{Flatten}[(\text{Pool} \circ \text{ReLU} \circ \text{Conv})^{L_{\text{CNN}}}(\text{ROI}_{v_i})]$$

Edge-aware attention, per layer and head

$$z_{ij} = \text{LeakyReLU}[\mathbf{a}^\top (\mathbf{W}_v \mathbf{h}_{v_i} \parallel \mathbf{W}_v \mathbf{h}_{v_j} \parallel \mathbf{W}_e \mathbf{h}_{e_{ij}})]$$

Edge prediction

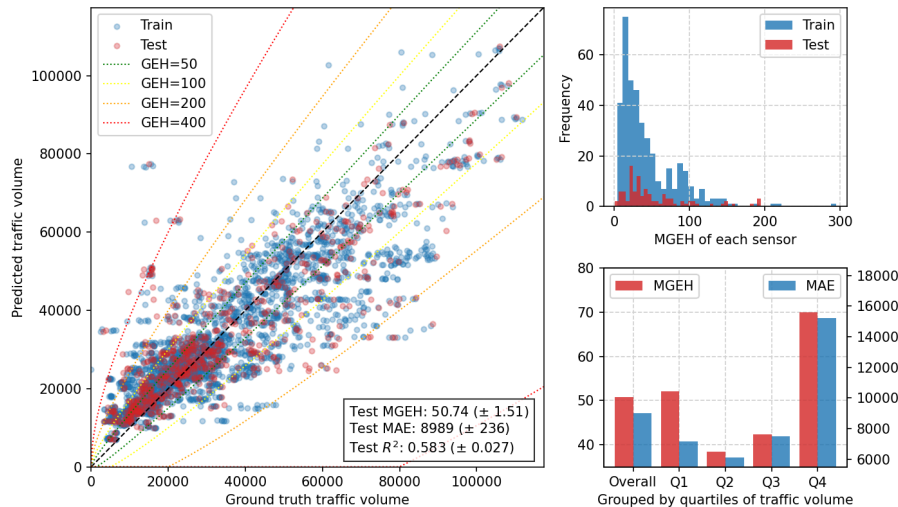
$$\hat{y}_{t,e} = \text{MLP}_{\text{pred}}(\mathbf{h}_{v_i}^{(L)} \parallel \mathbf{h}_{v_j}^{(L)} \parallel \mathbf{h}_e)$$

- Loss: mean GEH over training edges. Each batch holds the whole network for one year.
- 25 km region of interest, three convolutional layers (16, 32, 64 channels), five GAT layers, four heads, $d = 16$.

Mukara: prediction and spatial evaluation

Model	Random CV (five folds)			Spatial CV (nine regions)		
	MGEH	MAE	R^2	MGEH	MAE	R^2
Global mean	102.31	19,708	0	105.27	20,413	< 0
Gravity model	84.23	14,836	0.342	93.18	16,925	0.201
Ridge regression	75.64	12,942	0.463	86.72	15,109	0.302
Random forest	71.38	12,319	0.504	81.54	14,216	0.361
Mukara	50.74	8,989	0.583	57.63	9,955	0.521

Means across folds (Table 4.7). MAE in vehicles per day.

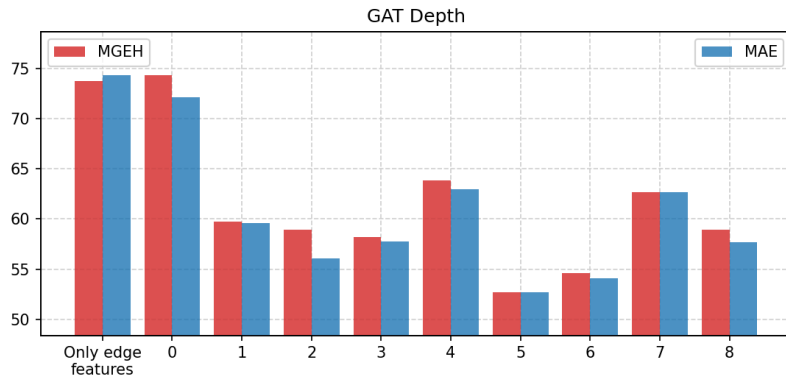


Predicted against observed volume, first CV fold (Figure 4.13)

- R^2 falls only from 0.583 to 0.521 when whole regions are withheld. Every baseline deteriorates more.
- MAE is 26.6% of the mean observed volume of 33,735 vehicles per day.
- Errors are largest at the lowest and highest volumes. Error maps show local over- and under-prediction without a regional pattern.
- Regional totals under spatial CV lie within $\pm 6\%$ of observed totals, the national total within 0.6%.

Mukara: what contributes to prediction

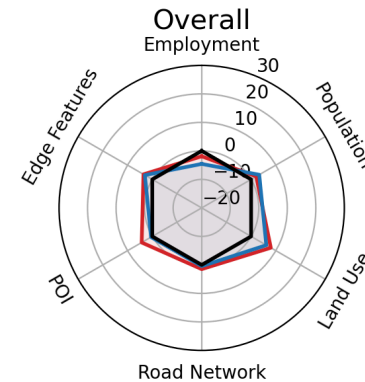
Architecture



Test MGEH and MAE by GAT depth (Figure 4.12)

- Edge features alone carry limited information. Endpoint embeddings add local context.
- Full graph propagation improves further. Five or six GAT layers perform best, and deeper models deteriorate.
- Ridge and random forest under spatial CV perform like the variant without GAT layers, so the gain is consistent with multi-hop propagation.

Feature groups



Change in MGEH (red) and MAE (blue) when one group is removed, % (Figure 4.16)

- Removing land use causes the largest increase in error.
- Stratified employment improves on aggregate employment. Removing employment reduces error in some settings, which indicates overlap with other inputs.
- Robust under spatial CV (R^2): 2019 OSM snapshot 0.517, all days 0.533, MSE, MAE or Huber loss 0.519–0.523 (Backup).

From Mukara to DeepDemand

Scope of the Mukara result

- Predictive, not causal. The feature analyses measure predictive contribution only.
- 498 monitored segments. Repeated years add examples but not locations.
- One 2023 OSM snapshot serves all study years.
- Transfer tested within England only. No temporal extrapolation, congestion or mode-shift response.
- A heteroscedastic Gaussian extension trained less stably and raised MAE from about 9,000 to about 12,000 vehicles per day, so point prediction is kept.

Representational limits and the DeepDemand response

Limit of Mukara	DeepDemand
Long trunk-road connections between major junctions hide variation near junctions and slip roads	Detailed national driving network with 5,088 target edges
Square raster windows: proximity is not accessibility, and rasterising changes the LSOA geography	LSOAs kept as units, reached through network travel time
Graph propagation does not identify which OD movements use the target edge	Explicit OD pairs, screened by shortest paths through the edge

Sections 4.5.4 and 5.2

CHAPTER 5

DeepDemand

RQ2 · Can a more explicit representation of OD relationships informed by travel-demand theory support prediction and interpretation at a finer road-segment resolution?

DeepDemand: theory-informed design

Two methodological developments:

- 1. Network preprocessing.** A competitive two-source Dijkstra search extracts origin and destination regions for each target edge. Shortest-path screening keeps the OD pairs whose shortest route uses the edge.
- 2. Differentiable model.** Learns a non-negative OD interaction score and a travel-time attenuation for every screened pair, and sums their contributions into a link volume.

Trained end to end on observed link volumes. No observed OD flows are used, so the learned OD scores are model-derived quantities, not trip counts.

Travel-demand concept	DeepDemand component
Trip generation and attraction	Separate origin and destination encoders f_O, f_D
Spatial interaction	OD interaction score s_{od} from f_{OD}
Deterrence function	Learned travel-time attenuation p_{od} from f_t
Traffic assignment	Shortest-path screening, then summation over the screened pairs $\mathcal{P}_{e^*}$. Assignment-like, not an equilibrium.



Framework stages (Figure 5.1)

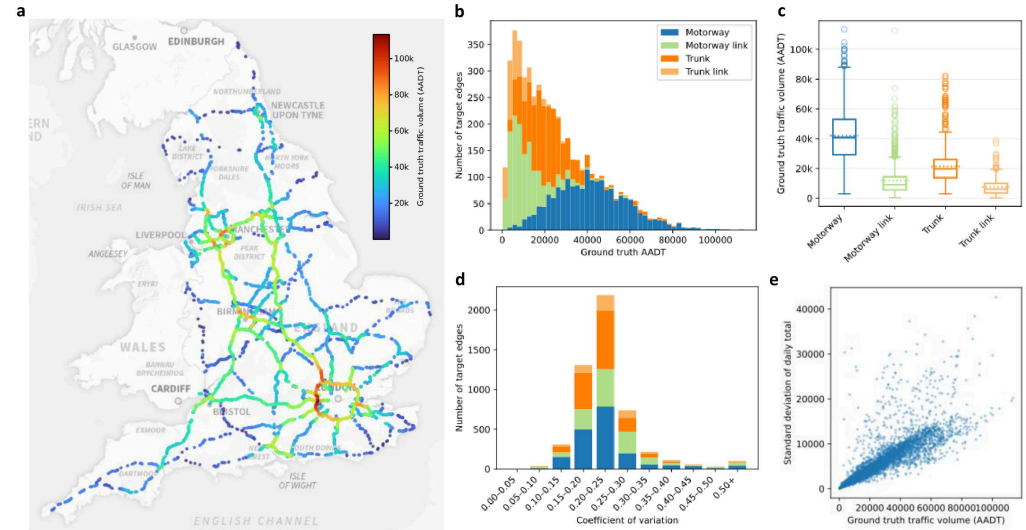
Design decisions from methodological development

Alternative explored	Observation	Final choice
Keep only major road classes	Truncated connections at junctions and slip roads, distorted travel-time surfaces	Full drivable network
Raw OSM segments as targets	Targets depend on digitisation and duplicate one another	Simplified graph between junctions
GNN on the full national graph	Millions of nodes for few targets, deep propagation, oversmoothing	Target-specific computation
Local subgraphs, shortest-path-tree aggregation	Reachability does not show that travel between two areas uses the target edge	Explicit OD pairs with route screening
Independent origin and destination searches	The two regions overlap as the horizon grows	Competitive two-source expansion
Learned spatial clustering	Unstable across runs, no consistent gain	Network accessibility as spatial structure
Softmax attention over OD pairs	The weight of a pair depends on the other pairs of the edge	Independent attenuation per pair
Relaxed screening (within $1.2 \times$ shortest time)	Many weakly related pairs, especially around slip roads	Exact shortest-path criterion
MLP aggregation	Unstable optimisation. A learned power S^a converged to a of about 0.41	Fixed square root, $\sqrt{S}$

Section 5.3.2. The findings concern the tested designs under link-count supervision and do not rule out other GNN formulations.

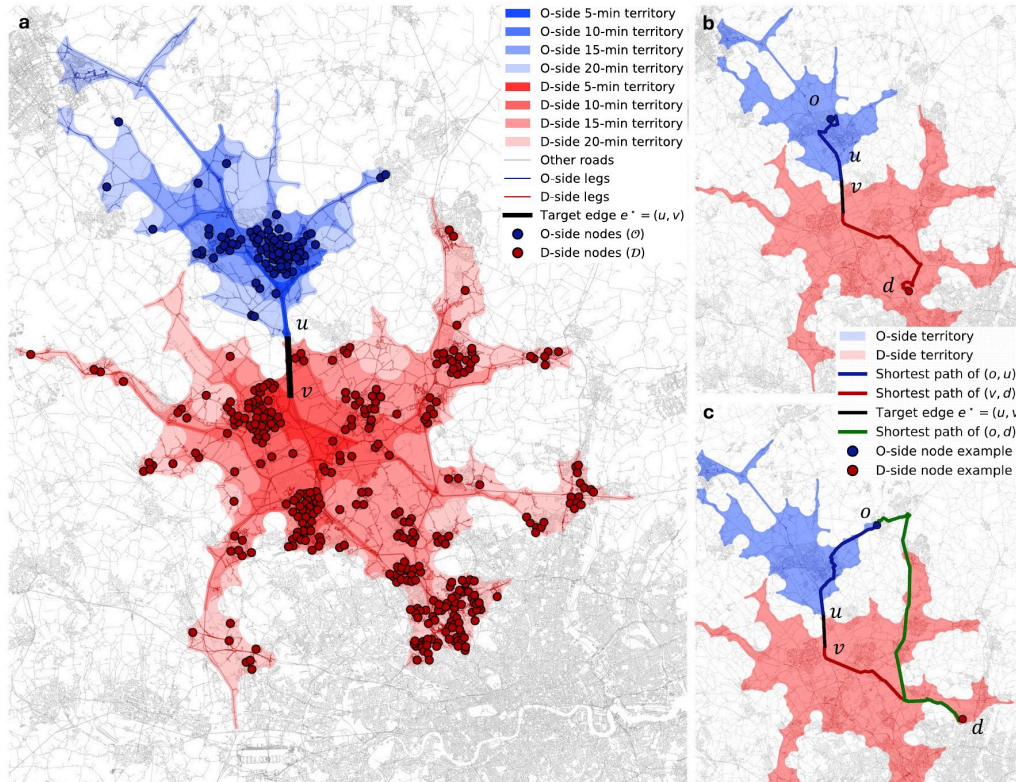
Network, features and targets

- **Driving network.** OpenStreetMap Great Britain, 1 January 2025. All drivable roads kept, then simplified to about 6.3 million nodes and 7.2 million directed edges. Free-flow travel time from the posted speed limit, or a road-class fallback speed.
- **LSOA features.** 121 variables on population, employment, households and car availability, deprivation, land use and POIs, standardised and reduced to 64 principal components. Each LSOA is attached to the node nearest its population-weighted centroid.
- **Targets.** TRIS sensors matched within 10 m to motorway, trunk and link edges. Annual AADT for 2017–2024 from sensor-years with at least 200 valid observations. Median across sensors, then across years.
- **5,088 target edges.** Mean AADT 25,243, standard deviation 18,893, maximum 113,436 vehicles per day.



Observed AADT: map, distributions by road type and day-to-day variability (Figure 5.5)

Target-specific OD extraction and screening



(a) Competitive expansion, shown with a 20-minute cutoff. (b) A retained pair. (c) An excluded pair (Figure 5.7)

- **Competitive two-source Dijkstra.** From $e^* = (u, v)$, search backward from u for origins and forward from v for destinations with one shared priority queue. The first search to reach a node claims it, so $\mathcal{O}_{e^*}$ and $\mathcal{D}_{e^*}$ do not overlap.
- **Cutoff** $\tau_{\max} = 60$ min on each side. In the NTS 2017–2024 (985,676 car trips), 3.7% of weighted car trips exceed 60 minutes.
- **Screening.** Keep (o, d) only if a shortest route passes through the edge:

$$\tau_O(o) + \tau_{e^*} + \tau_D(d) = \tau_{OD}(o, d)$$

- **National scale.** Compressed sparse row network (about 187 MB), compiled Dijkstra kernels, cached shortest-path trees and parallel workers. All 5,088 edges in about 33 minutes, 1.64 billion screened pairs across target edges.

Differentiable demand-to-flow model

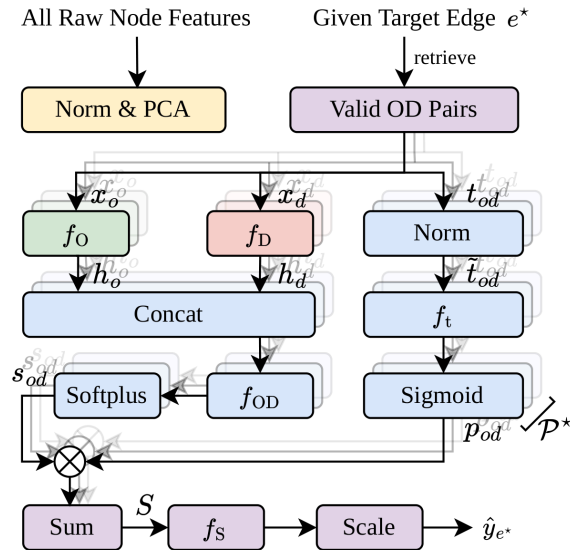


Figure 5.8

$$\mathbf{h}_o = f_O(\mathbf{x}_o^{(v)}), \quad \mathbf{h}_d = f_D(\mathbf{x}_d^{(v)})$$

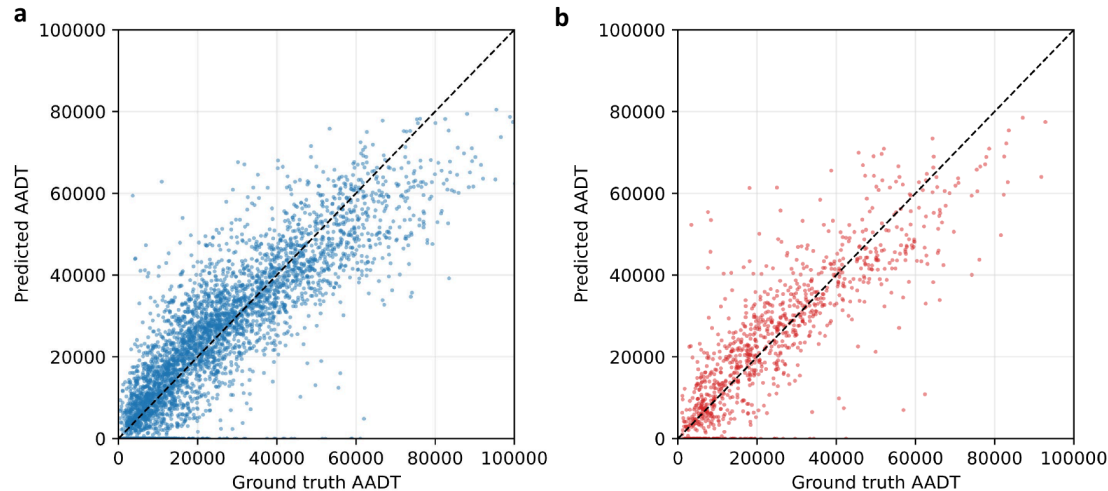
$$s_{od} = \text{Softplus}(f_{OD}([\mathbf{h}_o \parallel \mathbf{h}_d])) \geq 0$$

$$p_{od} = \sigma(f_t(\tilde{\tau}_{od})) \in (0, 1), \quad \tilde{\tau}_{od} = \frac{\tau_{OD}(o, d) - \bar{\tau}}{s_\tau}$$

$$S_{e^*} = \sum_{(o,d) \in \mathcal{P}_{e^*}} s_{od} p_{od}, \quad \hat{y}_{e^*} = \gamma \sqrt{S_{e^*}}$$

- f_O , f_D , f_{OD} and f_t are small MLPs trained jointly. $\bar{\tau} = 3600$ s, $s_\tau = 1000$ s, $\gamma = 100$.
- Each pair contributes independently. There is no softmax over pairs.
- Mean squared error loss. AdamW with staged learning rates 3×10^{-3} , 10^{-3} and 3×10^{-4} , early stopping on validation MGEH.

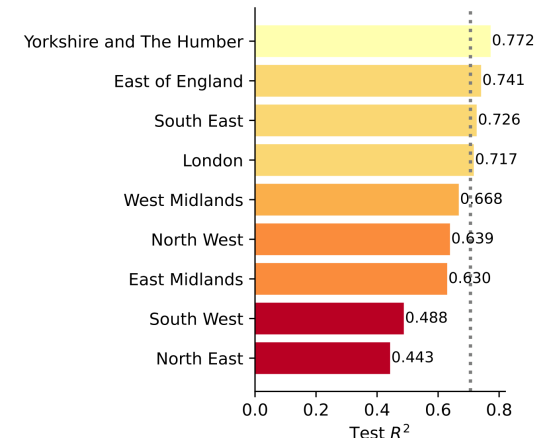
Predictive performance and spatial generalisation



Predicted against observed AADT, training (a) and test (b) edges, first random-CV fold (Figure 5.9)

- Training and test performance are similar, which indicates limited overfitting.
- Accuracy is similar across motorway, trunk and link roads, so nearby carriageways and slip roads with very different volumes are distinguished.

	Random CV	Spatial CV
Test R^2	0.706	0.647
MAE (veh/day)	7,567	7,885
wMAPE	30.0%	31.1%
MGEH	56.08	58.00



Test R^2 by held-out region, 0.443 to 0.772. Dotted line: random-CV mean (Figure 5.11)

Baseline comparison: representation matters

Model	Input	Random CV			Spatial CV		
		MGEH	MAE	R^2	MGEH	MAE	R^2
Mean	none	96.77	15,171	−0.001	97.45	15,402	−0.097
Linear regression	OD-structured	89.69	14,350	−0.003	93.15	14,956	−0.119
Ridge regression	OD-structured	89.38	14,141	0.062	93.15	14,876	−0.094
Gravity (log-linear)	screened pairs	65.46	9,379	0.546	66.50	9,521	0.486
Random forest	pooled	90.34	14,195	0.113	92.03	14,534	0.008
Random forest	OD-structured	57.68	8,659	0.623	66.37	10,318	0.445
XGBoost	pooled	90.02	14,141	0.115	92.04	14,514	0.003
XGBoost	OD-structured	53.23	7,975	0.656	63.16	9,776	0.458
LightGBM	pooled	89.57	14,042	0.130	91.91	14,493	0.005
LightGBM	OD-structured	51.14	7,652	0.678	61.26	9,487	0.481
GAT	local graph	72.39	11,102	0.402	74.35	11,552	0.319
DeepDemand	screened pairs	56.08	7,567	0.706	58.00	7,885	0.647

Means across folds on identical partitions (Table 5.4). Pooled: one catchment average, without origin–destination separation or screening.

- The same tree learners rise from R^2 0.11–0.13 with pooled features to 0.62–0.68 with OD-structured features. The OD representation carries much of the signal.
- The trees fit training edges almost exactly (training R^2 0.93–0.98) and lose the most when whole regions are held out. DeepDemand: training 0.73, test 0.706 and 0.647.
- The gravity model is restrictive but transfers stably, from 0.546 to 0.486.
- A GAT on the unscreened local graph, as tested, does not recover this structure.
- Boosted trees reach a lower MGEH under random CV, so the metrics do not rank the models identically.

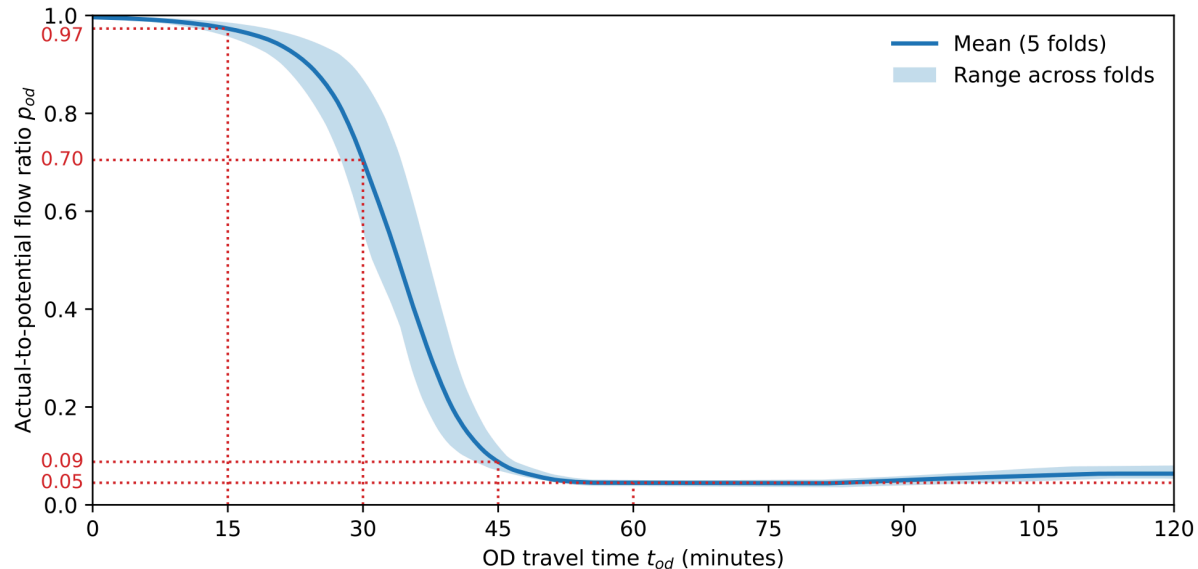
Component ablation

Variant (one component replaced)	Random CV		Spatial CV	
	MGEH	R^2 (change)	MGEH	R^2 (change)
DeepDemand, full model	56.08	0.706	58.00	0.647
No OD pair screening	57.86	0.670 (−0.036)	60.74	0.596 (−0.051)
No competitive partition and no screening	60.52	0.639 (−0.067)	63.55	0.548 (−0.099)
Shared origin and destination encoder	56.13	0.709 (+0.003)	57.75	0.645 (−0.002)
Constant travel-time attenuation	60.62	0.659 (−0.047)	66.46	0.503 (−0.144)
Linear aggregation instead of $\sqrt{S}$	69.50	0.624 (−0.081)	76.03	0.445 (−0.202)

Table 5.6. Each variant is retrained under the same protocol.

- Competitive partitioning and screening make separate contributions.
- Learned attenuation and the sublinear readout matter most, and their removal costs more under spatial CV. The structure supports geographical transfer as well as fit.
- Separate origin and destination encoders do not improve accuracy. A shared encoder is a legitimate simplification. The separation is kept because it organises the model and its interpretation.

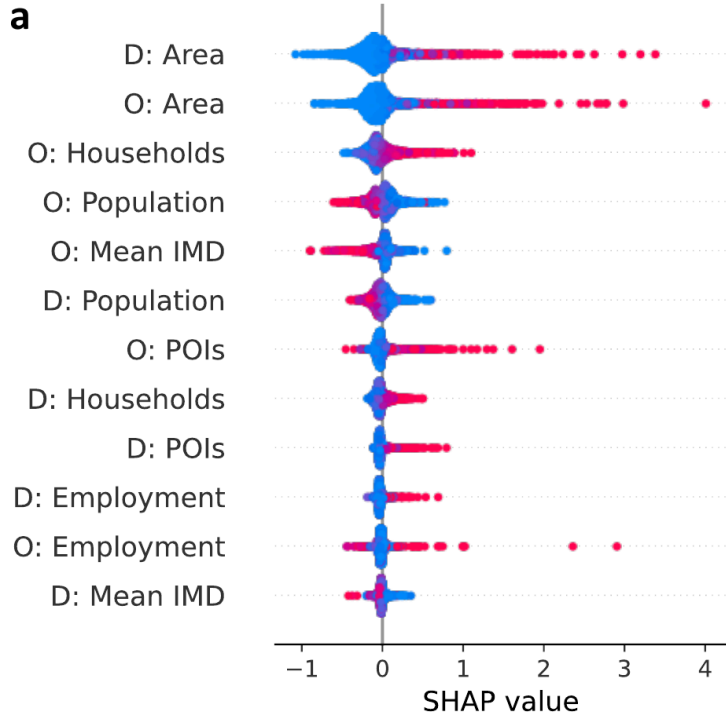
Interpretation: learned travel-time attenuation



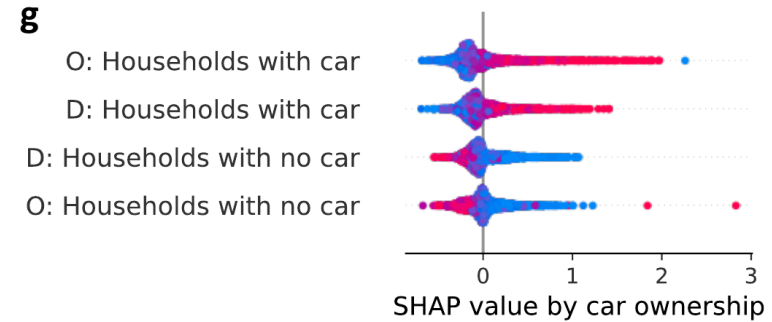
Attenuation p_{od} against OD travel time: mean over random-CV folds and range across folds (Figure 5.13)

- Nearly flat below about 15 minutes, then falls from about 0.97 to 0.09 between 15 and 45 minutes.
- Plateaus near 0.05 beyond 45 to 50 minutes and rises slightly to about 0.07 at 90 to 120 minutes.
- An MLP outperformed linear-logit and exponential-decay forms for f_t .
- Narrow range across folds. Spatial-CV curves share the shape, with modest shifts between held-out regions (Backup).
- Estimated from link counts only. It attenuates latent OD scores and is not an independently measured behavioural response.

Interpretation: area characteristics and OD scores



(a) Feature domains at origin (O) and destination (D). Red: high feature value (Figure 5.15)

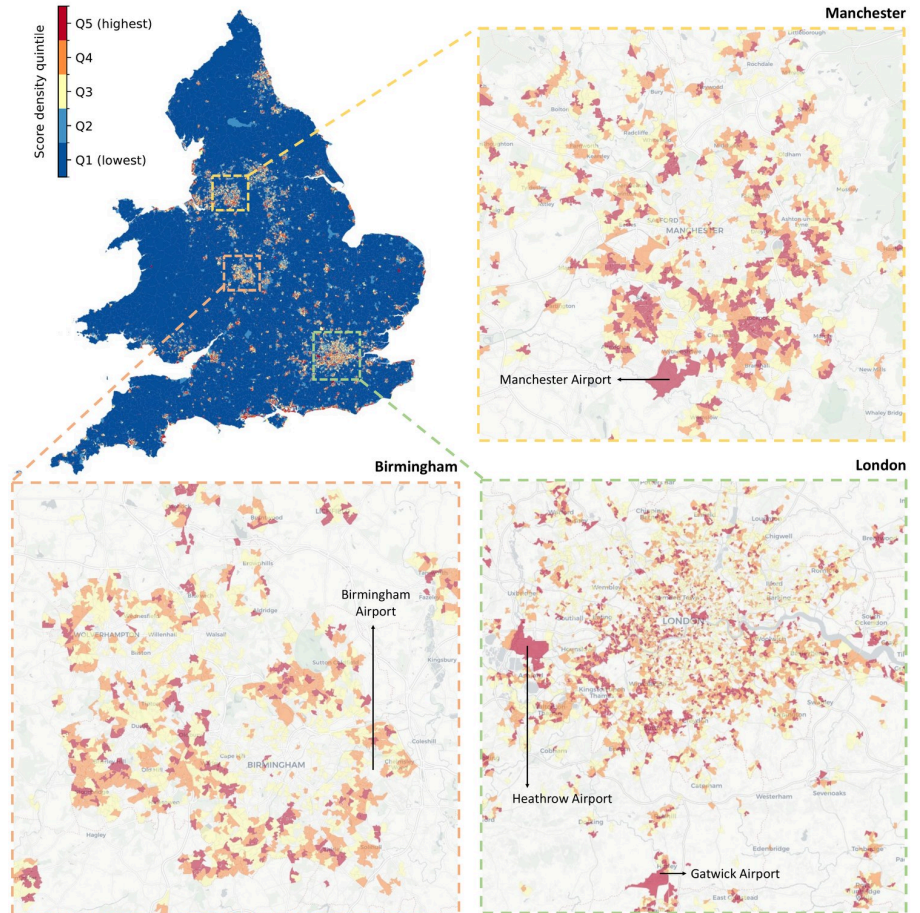


(g) Household car availability

- Random-forest surrogates, one per feature group, fitted to $\log s_{od}$ for 50,000 pairs sampled from 107,317,306 unique screened pairs. SHAP values on the raw LSOA variables.
- Larger LSOA area goes with higher scores, consistent with fewer zonal alternatives per pair. More households with cars, more residential and commercial land and more transport POIs also go with higher scores.
- More residents aged 16–24 go with lower scores, and more residents aged 50 and over with higher scores.
- These are associations with a fitted quantity, not causal effects and not the neural model's internal attribution.

Interpretation: origin and destination potentials

- Destination potential of an LSOA: its mean OD score when acting as destination, per unit area. Origin potential is defined in the same way.
- Destination potential is polycentric. The City of London, Canary Wharf, the Thames corridor, Heathrow, Gatwick and Manchester Airport stand out. Birmingham Airport is weaker.
- Origin potential is more diffuse and extends into suburban and peri-urban areas.
- The model saw no OD flows and no generation or attraction totals. The maps are hypotheses to test against such data, not identified OD estimates.

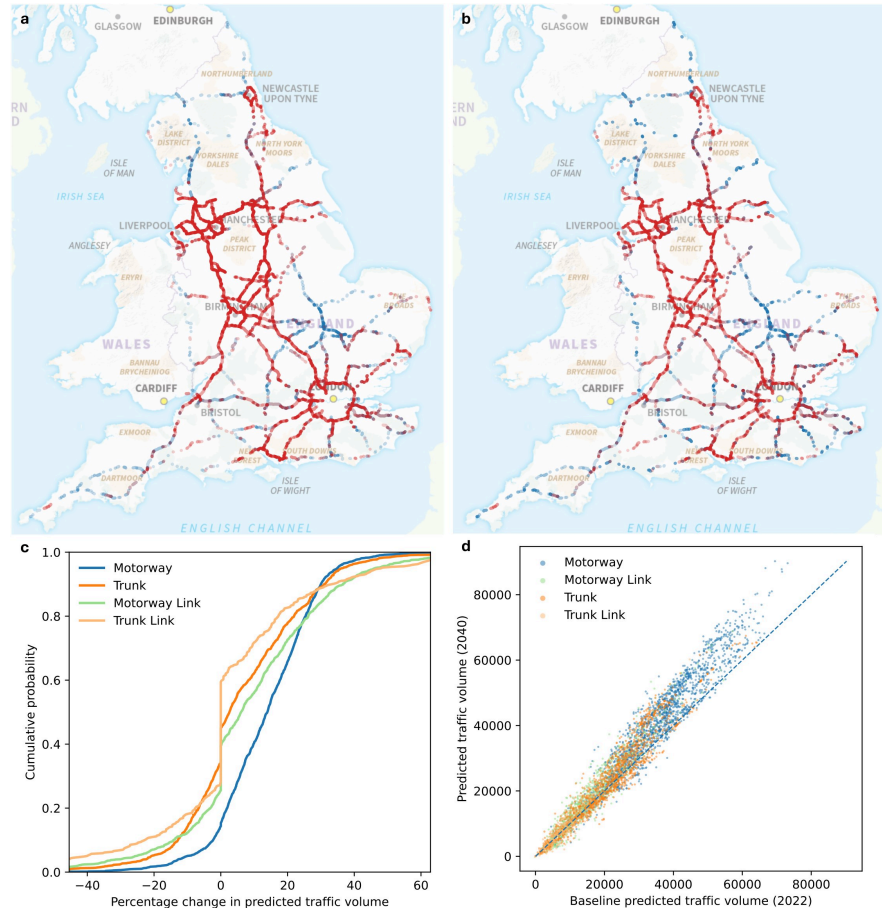


Destination potential by quintile, with Manchester, Birmingham and London (Figure 5.21)

Conditional population-change scenario

- A variant retrained on the 20 inputs that have projections: land area, population density and 18 age–sex population counts.
- ONS 2040 population projections replace the inputs. Predictions are compared with the 2022 baseline of the same model. Network and fitted relationships stay fixed.
- Largest increases on motorways and corridors around and between London, Birmingham, Manchester and Leeds, and on orbital roads. Many rural and peripheral roads change little or decrease.

A conditional response of the fitted model, not a validated forecast. Employment, land use, policy, mode shift, congestion and capacity are not represented.



(a, b) Absolute and percentage change from 2022. (c) Change by road class. (d) Segment comparison (Figure 5.23)

Sensitivity analyses

Analysis	Variant	Random CV R^2	Spatial CV R^2
Multi-year target	Median across years (main)	0.706	0.647
	Mean across years	0.709	0.645
	Median excluding pandemic years	0.709	0.651
Input vintage	All inputs dated before 2017	0.713	0.648
OD catchment radius	30 min (0.17 billion pairs)	0.649	0.570
	60 min (1.64 billion pairs, main)	0.706	0.647
	90 min (4.90 billion pairs)	0.702	0.646

Tables 5.7 to 5.9. Each variant is retrained under the full protocol.

- Conclusions do not depend on how the multi-year target is summarised, or on using external inputs observed during or after the traffic period.
- Extending the catchment from 30 to 60 minutes raises R^2 by 0.057 and 0.077. Extending it to 90 minutes triples the pairs without gain, so one hour captures most of the available information.

DeepDemand: scope and limitations

Mode choice

Not represented. Car availability and accessibility carry some modal information, but modal shares and competing services are not modelled.

Static free-flow travel times

Extraction and screening ignore time of day, recurrent congestion and incidents. This suits a long-term AADT target but not time-varying route choice.

Assignment-like mapping

Screened contributions are summed. There is no equilibrium, capacity or congestion response, and equally short alternative routes are not split.

Spatial estimation only

No short-term variability and no predictive uncertainty. Temporal forecasting needs its own validation design.

Surrogate interpretation

SHAP describes associations between raw LSOA variables and fitted OD scores, not the model's internal use of principal components.

Freight beyond one hour

The NTS covers personal travel only, and mean HGV length of haul is about 105 km. Cars are about 74% of vehicle miles and HGVs about 6%, so the catchment is supported for cars but not for freight.

CHAPTER 6

Conclusions

Findings · contributions · planning relevance · further work

Answers to the research questions

RQ1

Yes, within the study setting. Mukara estimates weekday traffic on 498 trunk-road segments from external characteristics and network structure, with test R^2 0.583 under random CV and 0.521 under spatial CV, ahead of the gravity, ridge and random-forest baselines. Spatial context and network propagation both contribute.

RQ2

Yes, for specific structural choices. DeepDemand reaches test R^2 0.706 and 0.647 on 5,088 edges, with the lowest MAE and wMAPE of all models under its protocol. OD extraction and screening lift the same tree learners far above their pooled versions. Ablations support partitioning, screening, learned attenuation and sublinear aggregation, but not separate encoders.

RQ3

Both models transfer across English regions (0.583 to 0.521, and 0.706 to 0.647), while high-capacity trees lose more. DeepDemand exposes a stable attenuation curve, feature associations and spatial potentials. These are model-derived quantities that suggest hypotheses, not identified behavioural parameters.

Synthesis: from Mukara to DeepDemand

	Mukara	DeepDemand
Prediction unit	498 directed trunk-road segments between major junctions	5,088 monitored edges on the detailed driving network
Traffic target	Annual mean weekday daily traffic, 2015–2022	Median AADT across valid years, 2017–2024
External representation	Rasterised population, employment, land use, road context and POIs	LSOA demographic, socioeconomic, land-use and POI characteristics
Network representation	Compact trunk-road graph with graph-attention propagation	Great Britain driving network with target-specific OD extraction and screening
Demand representation	Mostly implicit in learned embeddings	Explicit screened OD pairs with learned scores and attenuation
Test R^2 , random / spatial	0.583 / 0.521	0.706 / 0.647
Structural finding	Spatial context and network propagation both contribute	Partitioning, screening, attenuation and sublinear aggregation support prediction and transfer
Inspectable quantities	Feature-group and architectural contributions	Travel-time attenuation, OD scores, origin and destination potentials

After Table 6.1. The R^2 values are not directly comparable: units, targets, periods and protocols differ.

Contributions

1 • Observation-supervised external-input estimation

Two models learn link-level traffic from observed counts, with only external socioeconomic and network information at prediction: no traffic-state input and no observed OD matrix. The claim is this conjunction at national scale, not the first use of external variables.

2 • Explicit test of geographical generalisation

Region-held-out CV alongside random CV in both studies. It separates models that transfer from models that only fit, and it bounds the claims: it is not future-year forecasting, transfer to another country or intervention analysis.

3 • Theory-informed OD representation at national scale

Competitive two-source Dijkstra, shortest-path screening, learned interaction and travel-time attenuation. Routing on 6.3 million nodes becomes a reusable preprocessing step, and the representation is separable from the neural architecture.

Observed link counts can supervise models that learn demand-to-flow relationships from external characteristics and network structure. How that relationship is represented matters for accuracy, transfer and interpretation, beyond model capacity alone.

Planning relevance and scope

Supports

- Estimates at unmonitored roads where compatible socioeconomic and network data exist.
- Network screening, preliminary demand assessment and targeting of new monitoring.
- Conditional scenario exploration under stated assumptions, such as the 2040 population projections.
- Exploratory analysis of learned demand structure: attenuation, OD scores and potentials.

Does not establish

- Validated long-term traffic forecasts.
- Causal effects of housing, pricing or infrastructure interventions.
- Mode choice, congested equilibrium or time-varying route choice.
- Transfer to other countries, dense urban networks or different road hierarchies.
- Precise engineering estimates without local calibration or measurement.

Traffic observations still matter. Performance depends on the representativeness and quality of the training counts, and a new setting needs validation and a check that it lies within the training domain.

Directions for further work

Coverage and inputs

- Automatic sensor matching for more targets
- Time-varying land use, capacity and junction detail
- Finer zones, which need faster screening

Demand structure

- Explicit mode choice
- Multi-path, equilibrium or dynamic assignment
- Time-dependent costs and freight-specific catchments
- Synthetic benchmarks with known ground truth

Architecture and scaling

- Other learners on the same OD representation
- Simpler variants such as a shared encoder
- Faster routing and screening for finer zones

Temporal validation

- Rolling-origin tests on genuinely later data
- Joint projections of population, employment and transport supply

Predictive uncertainty

- Ensembles and Bayesian methods
- Assessment by calibration, coverage and interval width

Transfer and causality

- Other countries and few-shot calibration
- Causal designs for intervention questions

Thank you

Questions and discussion

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APPENDIX

Backup slides

Four-step model: key relationships

Distribution

$$\begin{aligned}T_{ij} &= A_i O_i B_j D_j f(c_{ij}) \\ A_i &= \left[\sum_j B_j D_j f(c_{ij}) \right]^{-1} \\ B_j &= \left[\sum_i A_i O_i f(c_{ij}) \right]^{-1}\end{aligned}$$

Maximising $\log W$ with $W = T! / \prod_{ij} T_{ij}!$, subject to the margins and the total cost $C = \sum_{ij} T_{ij} c_{ij}$, gives $f(c_{ij}) = \exp(-\beta c_{ij})$ (Wilson).

Mode choice

$$P_{ij}^k = \frac{\exp(V_{ij}^k)}{\sum_{k' \in \mathcal{K}_{ij}} \exp(V_{ij}^{k'})}, \quad T_{ij}^k = P_{ij}^k T_{ij}$$

Assignment

$$\begin{aligned}\sum_{r \in \mathcal{R}_{ij}} f_{ijr} &= q_{ij}, & v_a &= \sum_{ij} \sum_{r \in \mathcal{R}_{ij}} \delta_{a,ijr} f_{ijr} \\ \min_{\mathbf{f}} \sum_a \int_0^{v_a} t_a(x) dx\end{aligned}$$

- User equilibrium: all used routes of an OD pair have equal, minimal cost. Link costs follow a volume–delay function such as BPR.
- Many OD and route-flow patterns give the same link totals, so counts alone do not identify a unique OD matrix.

GEH on daily volumes

- Reference values of 5 and 10 come from hourly assignment validation. Multiplying both flows by c multiplies GEH by $\sqrt{c}$.
- If daily traffic is about ten times peak-hour traffic, illustrative daily levels are about 16 and 32. They are not acceptance thresholds.
- DeepDemand: 19.5% of target edges below 16 and 37.0% below 32 under random CV, 18.3% and 36.3% under spatial CV.
- Mukara: 18% of sensors below 16 and 49% below 32 in the random-CV test sets. Sensor-level MGEH quartiles 23.88, 32.10 and 72.51.
- Many individual links keep substantial error even where network-level R^2 , MAE and wMAPE are useful.

Relation to R^2 . With small relative errors and a volume-weighted RMS GEH G ,

$$R^2 \approx 1 - G^2 \frac{\bar{y}}{\sigma_y^2}$$

For this network $\bar{y} = 25,243$ and $\sigma_y = 18,893$, so $G = 16$ corresponds to $R^2 \approx 0.982$ and $G = 32$ to $R^2 \approx 0.928$.

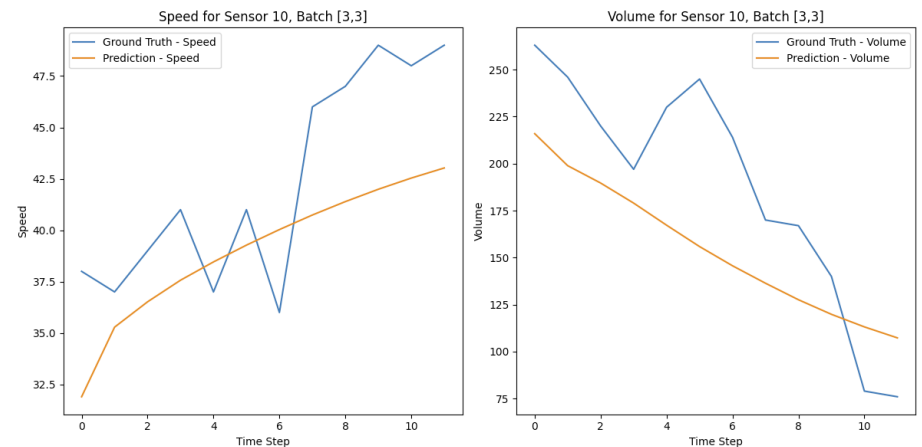
RMS GEH is not MGEH, so there is no general conversion from the reported MGEH to R^2 (Appendix A.3).

ST-MetaNet case study: setup and errors

- 498 monitored segments, 35,040 timestamps from 2022, twelve 15-minute intervals in and out.
- Inputs: speed and volume, missing-data masks, calendar information, road and junction metadata.
- Chronological 70/10/20 split. Errors after three epochs:

Variable	Metric	Training	Validation
Speed	MAE (km/h)	1.527	2.538
Speed	RMSE (km/h)	3.308	5.237
Volume	MAE (veh/15 min)	25.74	68.85
Volume	RMSE (veh/15 min)	41.65	126.28

Table 2.2. Appendix B gives the architecture and training details.



Blind extrapolation: smooth continuation of the recent trend (Figure 2.5)

Follow-up: 30-minute averages and a twelve-to-one structure (six hours of history, next half hour), shown for 4 to 10 November 2019.

Mukara: representative sensor selection

Step	Remaining	Excluded
Candidate sensors mapped to the 498 segments	3,984	–
Topology filter: slip roads, ramps, junction influence zones	2,815	1,169
One sensor per segment, availability first	498	2,317
Fallback: opposite-direction proxy	3 segments	–

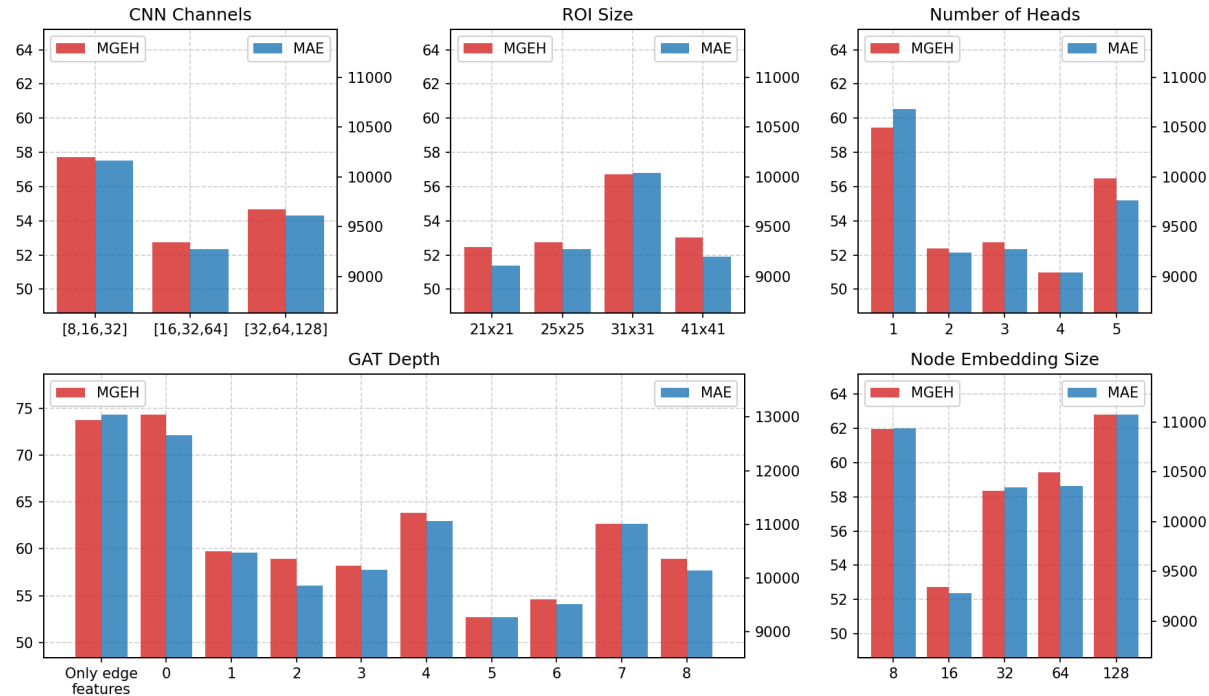
Table 4.2

Characteristic	Included (498)	Discarded (3,486)
Motorway share	60.2%	57.1%
Mainline position	100%	66.8%
London share	6.4%	14.9%
Mean volume (SD), veh/day	33,735 (21,999)	30,151 (27,358)
Temporal coverage (SD)	0.86 (0.10)	0.31 (0.24)

Table 4.3

- Selection favours mainline sensors with high completeness. The national traffic pattern is retained (Figure 4.4), but a comparison of means cannot establish representativeness across the whole volume distribution.

Mukara: architecture tuning



Test MGEH and MAE by configuration (Figure 4.12)

- Best individual settings: CNN channels (16, 32, 64), a 21×21 km region of interest, four heads, five GAT layers and embedding size 16.
- The final configuration adopts four heads and keeps the default 25 km region of interest. Loss reaches its minimum near epoch 27, and early stopping with a learning-rate schedule gave a similar best result.

Mukara: robustness checks under spatial CV

Check	Variant	MGEH	MAE	R^2
Main model	2023 OSM, weekdays, MGEH loss	57.63	9,955	0.521
OSM vintage	1 January 2019 snapshot	58.14	10,021	0.517
Target days	All days, including weekends and holidays	56.48	9,734	0.533
Training loss	MSE	58.12	9,901	0.523
	MAE	58.64	9,843	0.519
	Huber	57.91	9,886	0.522

Tables 4.8 to 4.10

MGEH training favours the lowest-volume quartile (MGEH 51.5 against 55.9 with MSE). MSE training favours the highest (67.2 against 69.8). The loss redistributes error rather than changing predictive capacity (Table 4.11).

Region	Observed	Predicted	Error
North East	0.728	0.756	+3.8%
North West	2.342	2.296	−2.0%
Yorkshire and The Humber	2.063	2.128	+3.2%
East Midlands	2.781	2.719	−2.2%
West Midlands	2.654	2.732	+2.9%
East of England	1.742	1.703	−2.2%
London	1.964	1.856	−5.5%
South East	1.897	1.842	−2.9%
South West	0.629	0.667	+6.0%
National	16.800	16.699	−0.6%

Sums of segment volumes in held-out regions, million vehicles per day (Table 4.13). Not unique trips, and errors can cancel.

DeepDemand: LSOA feature groups

Feature group	Stratification	Features	TDM relevance	Source
Area and population density	Land area, population density	2	Generation, distribution	ONS
Population	Total, sex (2), age groups (5), sex–age (10)	18	Generation	ONS
Employment	Total, work type (2), sector (18), work type–sector (36)	57	Attraction	ONS (BRES)
Households and cars	Total, car or van availability (2), composition (10), combinations (20)	33	Generation, mode choice	Census 2021
Deprivation	Income and employment domain scores, 2019	2	Generation, distribution	Deprivation data
Land-use area	Residential, commercial, industrial, retail	4	Generation, attraction, distribution	OSM
POIs	Transport, food, health, education, retail	5	Attraction, distribution	OSM

Table 5.1. 121 variables harmonised to 2021 LSOAs (2011-based sources converted with the ONS exact-fit lookup), standardised and reduced to 64 principal components.

DeepDemand: baseline construction

- **OD-structured features** (242): mean raw LSOA vector over the unique screened origins, concatenated with the mean over the unique screened destinations.
- **Pooled features** (121): mean over all LSOAs in the unscreened one-hour region, with no origin–destination separation and no screening.
- **Gravity:** $G_{e^*}(\lambda) = \sum_{(o,d) \in \mathcal{P}_{e^*}} M_o M_d \exp(-\lambda \tau_{OD}(o, d))$, with masses from population plus employment and $\log y = a_0 + a_1 \log(G + \epsilon)$. λ is chosen on validation data.
- **GAT:** two-hub local graph. Candidate origins link to u and destinations to v , with access and egress times as edge features, and one to six attention layers chosen on validation data. No screening.
- Hyperparameters of every model are selected on validation MGEH. Test data are used only for final evaluation.

Aspect	Mukara	DeepDemand
Targets	498 manually built segments	5,088 automatically matched edges
Years	2015–2022	2017–2024
Target	Annual weekday volume	Median AADT across years
Features	1 km rasters	121 LSOA variables, 64 components
Random CV	Five folds, four fitted and one tested	70/10/20 within five folds
Spatial CV	Nine regions	Nine regions

Table 5.3. Mukara is an external reference, not a controlled baseline.

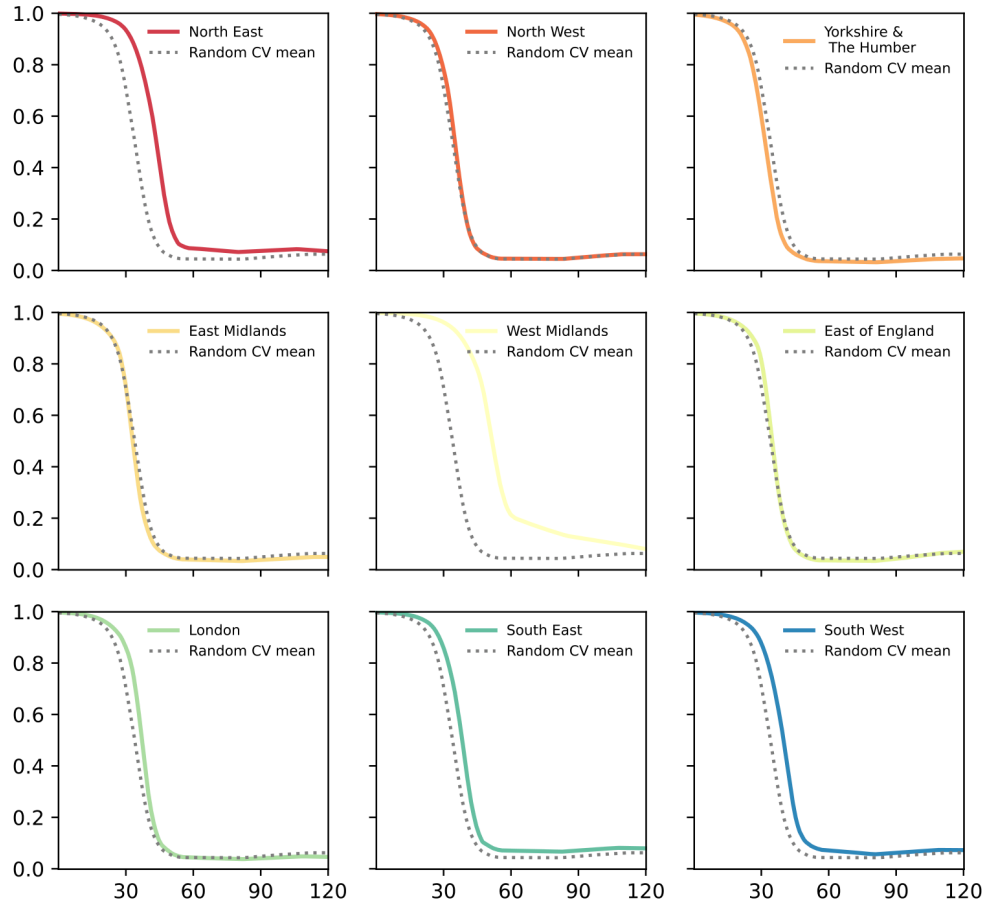
DeepDemand: training against test performance

Model	Random CV R^2		Spatial CV R^2	
	Training	Test	Training	Test
Gravity (log-linear)	0.546	0.546	0.546	0.486
Random forest (OD-structured)	0.931	0.623	0.934	0.445
XGBoost (OD-structured)	0.976	0.656	0.977	0.458
LightGBM (OD-structured)	0.973	0.678	0.973	0.481
GAT	0.567	0.402	0.592	0.319
DeepDemand	0.729	0.706	0.729	0.647

Tables 5.4 and 5.5, means across folds

The tree ensembles fit the training edges almost exactly but transfer worst. The train–test gap of DeepDemand is small under both schemes.

DeepDemand: attenuation by held-out region



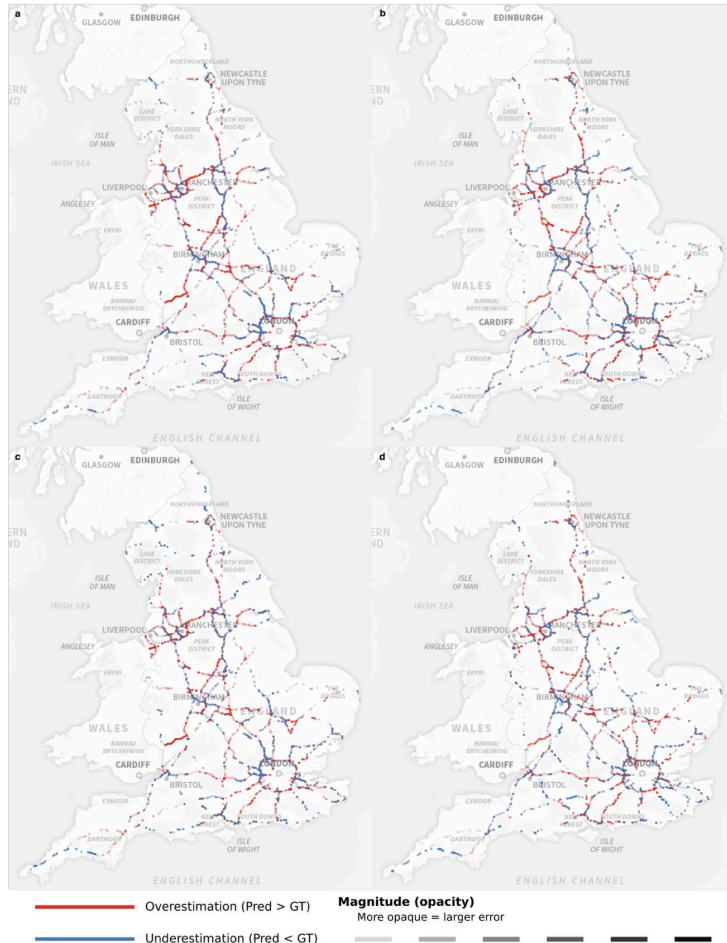
Spatial-CV attenuation curves against the random-CV mean (Figure 5.13)

- Every fold learns the same overall form, with modest shifts between held-out regions.
- The shifts suggest some geographical variation in the fitted sensitivity to travel time.
- They are properties of the fitted model, estimated from link counts, not measured behaviour.

DeepDemand: training settings and compute

- Random CV: 70/10/20 split of target edges in each fold. Spatial CV: one region for testing and 12.5% of the other regions' edges for validation.
- Edge-centric batches: eight target edges per pass, gradients accumulated over four passes, an effective batch of 32.
- AdamW, weight decay 10^{-4} , gradient-norm clipping at 5.
- Learning rates 3×10^{-3} , 10^{-3} and 3×10^{-4} . A stage ends when validation MGEH fails to improve by 0.1 over 15 evaluations, and its best checkpoint seeds the next stage.
- MLP widths: f_O and f_D [16, 16], f_{OD} [16, 8], f_t [16, 16].
- Workstation: Intel Core i7-13700F, 16 GB RAM, NVIDIA RTX 4060 Ti with 8 GB, PyTorch 2.5.1.
- Preprocessing of all 5,088 edges: about 33 minutes, done once and cached.
- Training: about 28 minutes per random-CV fold and 50 minutes per spatial-CV fold.
- Shortest-path trees are cached to $2\tau_{\max}$. A pair whose full journey through the edge exceeds this horizon can be missed, a small approximation.
- Data: doi.org/10.17863/CAM.127892. Code: github.com/yueli901/DeepDemand.

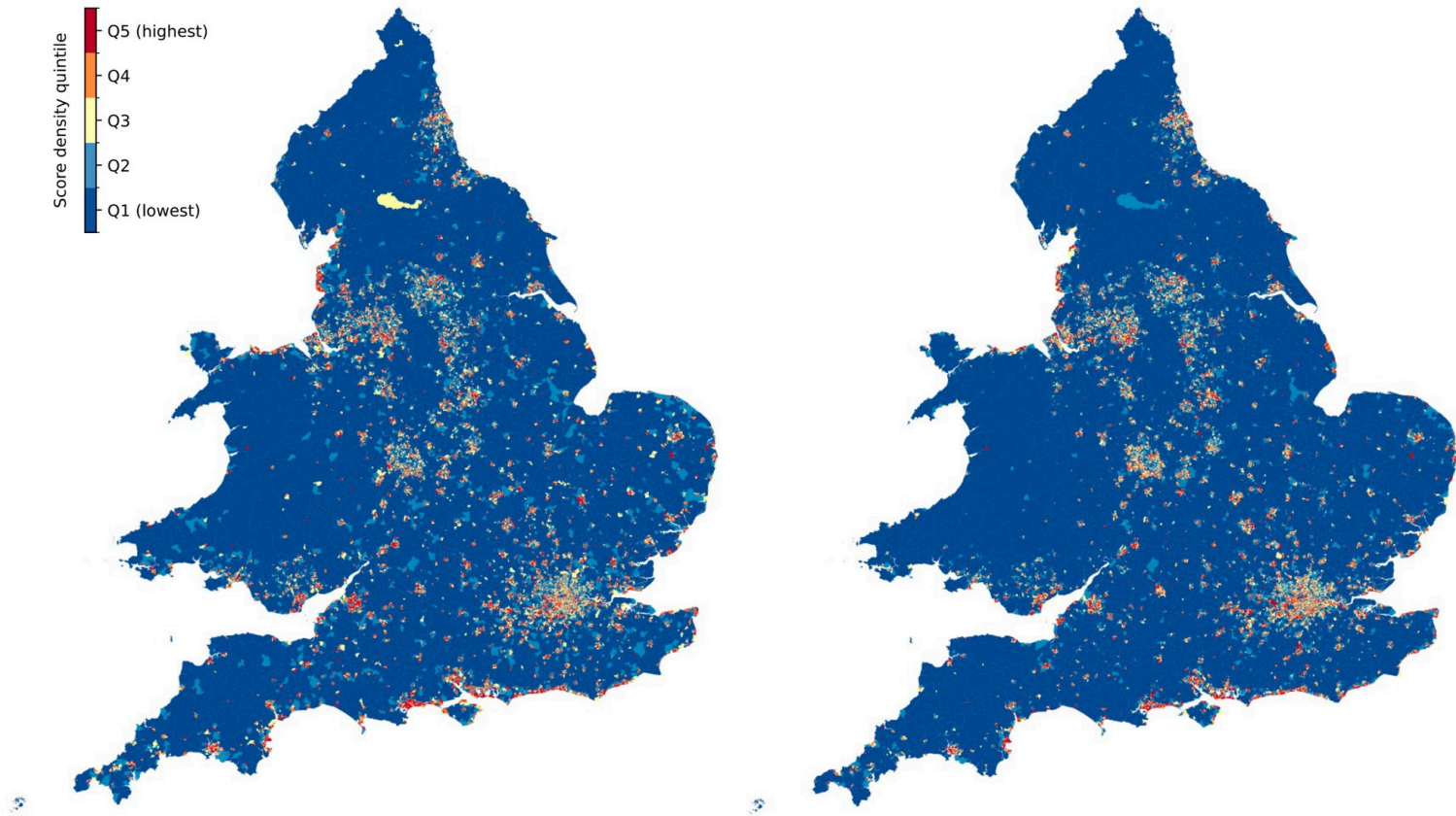
DeepDemand: spatial distribution of errors



- Errors are spread across the network without an obvious large-scale concentration or consistent directional pattern.
- Local over- and under-prediction remains. The absence of a visible pattern does not imply spatially independent residuals.
- Error varies little with the number of valid years or matched sensors per edge (Figure 5.10).

MAE and MGEH by direction. Red: overestimation. Blue: underestimation (Figure 5.12)

DeepDemand: origin and destination potentials



Origin potential (left) and destination potential (right), quintiles after standardisation by LSOA area (Figure 5.22)